SmolDocling:
Reproducing an Ultra-Compact VLM for End-to-End Multi-Modal Document Conversion

Phase 6 — Final Technical Report

Group ID: F26-04 | Track C — Research Paper Implementation

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Paper: Nassar et al., 2025 (arXiv:2503.11576)
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Abstract **3**
1. Project Summary **4**
2. Architecture and Methodology **5**
2.1 Paper Overview 5
2.2 Architecture Diagram 5
2.3 Methodology 6
3. Implementation Steps and Workflow **7**
3.1 Environment Setup 7
3.2 Data Pipeline (Phase 2) 7
3.3 Model Inference Pipeline (Phase 3) 7
3.4 Evaluation Pipeline (Phase 4) 8
3.5 Workflow Diagram 8
4. Replication Results vs. Reported Paper Results **9**
4.1 Im2LaTeX-230K — Equation Recognition (n=50) 9
4.2 DocLayNet v1.2 — Full-Page OCR (n=47) 9
4.3 PubTables-1M — Table Structure Recognition (n=100) 10
5. Challenges Faced During Implementation **11**
5.1 Environment Setup 11
5.2 Inference Behavior 11
5.3 Evaluation Pitfalls 11
6. Key Insights, Error Analysis, and Evaluation Summary **13**
6.1 Error Analysis 13
6.2 Bias and Fairness Analysis 14
6.3 Theoretical Insights 15
6.4 Evaluation Summary 15
7. Instructions to Run Code and Reproducibility **16**
7.1 Prerequisites 16
7.2 Base Models & Resources 16
7.3 Step-by-Step Reproduction 16
7.4 Repository Structure 17
8. References **18**

Abstract

This project is a reproduction and evaluation of an ultra-compact 256M parameter vision language model known as SmolDocling. SmolDocling is designed for end-to-end document understanding. This project's goal is to validate the paper's primary claim that a model designed for general purposes can outperform a highly domain-specific model when it comes to multi-modal document tasks.

For this evaluation, three datasets were used. Im2LaTex-230K was used to recognize equations, PubTables-1M is used for extraction of table structure and DocLayNet v1.2 was used for full page OCR. SmolDocling was compared with two baseline models, Nougat-base (350M) for OCR and equation related tasks and Microsoft table Transformer (TATR) for detecting table structures.

The methodology of this project involved the preprocessing of all datasets into unified DocTags/OTSL representations, usage of pretrained models for inference and performance evaluation using standard metrics i.e. Edit Distance, BLEU, METEOR, F1-score and structure only TEDS. Normalization techniques were highly considered to allow fair comparisons amongst models.

The result of this project depicts that SmolDocling can outperform Nougat on both tasks i.e. recognizing equations and full-page OCR while TATR outperforms SmolDocling for table structure tasks. Although SmolDocling is slightly weak in terms of structural accuracy, it still can extract both textual and structural content in one go.

To conclude, the project can reproduce the essential claims of the original paper and primarily confirms that compact vision-language models can achieve great performance across varying document understanding tasks.

1. Project Summary

Paper

SmolDocling: An Ultra-Compact Vision-Language Model for End-to-End Multi-Modal Document Conversion (Nassar et al., 2025)

Key Objectives

We reproduced results of the SmolDocling paper and compared them with domain specific baseline models. We performed our evaluation based on three tasks, equation recognition, table structure extraction and full-page OCR.

Models evaluated

1. SmolDocling-256M-preview (primary) — 256M parameter VLM built on SmolVLM backbone
2. Nougat-base (baseline) — 350M parameter document-to-Markdown model
3. TATR (Table Transformer, baseline) — dedicated table structure recognition model

Datasets

4. Im2LaTeX-230K — 238,329 equation image→LaTeX pairs (equation recognition)
5. PubTables-1M — 93,834 table structure annotations (table structure recognition)
6. DocLayNet v1.2 — 4,999 full document pages across 6 categories (full-page OCR)

Key Findings

SmolDocling was able to outperform Nougat for equation recognition by scoring an F1 score of 0.91 while Nougat's F1 score was 0.02. In reference to OCR tasks, Nougat scored an F1 of 0.37 while SmolDocling was able to score 0.66 in comparison. SmolDocling underperforms against TATR for table structure extraction tasks with a slightly less TEDS score of 0.866 compared to TATR's 0.942.

Our SmolDocling replication matches the paper's performance closely, it replicated the same results as the paper for Im2LaTex and PubTables-1M datasets but as the subset for DocLayNet used in our replication was not as diverse as used in the original paper, it was not able to replicate the exact results as shown in the paper.

Core Insight

A general purpose VLM like SmolDocling has the potential to replace highly specialized models as it shows competitive accuracy and versatility in its performance.

2. Architecture and Methodology

2.1 Paper Overview

SmolDocling is a 256M-parameter vision-language model designed for end-to-end multi-modal document conversion. Unlike specialist models that handle only one document element (e.g., tables or equations), SmolDocling processes entire document pages and outputs a unified structured representation called DocTags — a custom XML-like markup that encodes both layout coordinates and textual content in a single token stream.

The model is built on the SmolVLM backbone and follows an encoder-decoder architecture with three core stages. First, a Vision Encoder (SigLIP-SO400M) processes the input page image and produces a grid of visual embeddings that capture spatial and textual features from the document. Second, a Projection and Pooling module compresses these high-dimensional visual embeddings into a compact sequence of projected visual embeddings. This stage uses pixel-shuffle downsampling to reduce the token count by a factor of 9 (from roughly 1,521 image patches down to 169 tokens), which is critical for keeping the model small enough to run on constrained hardware. Third, these projected visual embeddings are concatenated with textual embeddings derived from the user's text prompt (e.g., "Convert this page to docling") and fed into a language model (LLM) decoder that autoregressively generates the DocTags output.

The LLM backbone is derived from SmolLM2-135M, a compact language model. To teach SmolDocling document-specific skills without catastrophic forgetting, the authors used a two-pronged training data strategy:

(a) rule-based instruction generation using Granite-3.1-2B-Instruct to create tasks like "Perform OCR at bbox" and "Identify page element type at bbox" from DocLayNet-PT pages, and

(b) co-training with the Cauldron conversation dataset to preserve general language understanding. Training was conducted on 64 NVIDIA A100 80GB GPUs for 4 epochs at approximately 38 hours per epoch.

The DocTags output format encodes each page element with its type tag (<text>, <table>, <formula>, <section_header>, etc.), bounding box coordinates quantized to a 500-point grid (<loc_0> through <loc_499>), and the element's textual content. For tables specifically, SmolDocling outputs OTSL (Optimized Table Structure Language), which uses tokens like <fcel>, <ecel>, <ched>, <rhed>, and <srow> to represent cell structure, spanning, and header roles. This unified output means a single forward pass can handle OCR, layout detection, table parsing, and equation recognition simultaneously.

2.2 Architecture Diagram

The figure below (reproduced from the paper) illustrates SmolDocling's end-to-end pipeline:

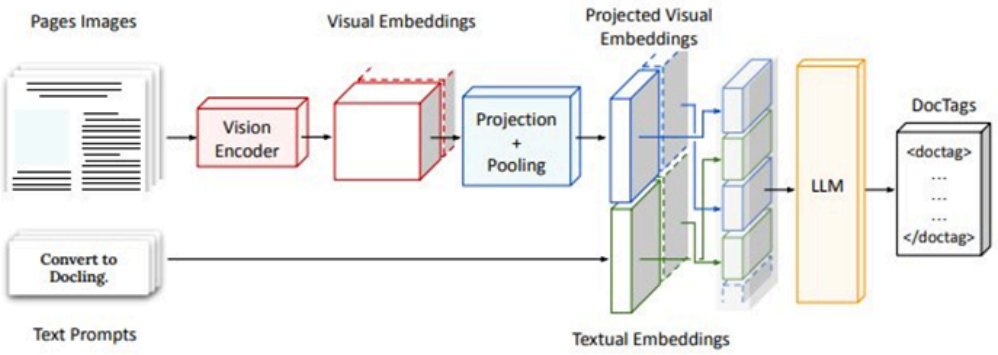

Figure 1: SmolDocling architecture (source: Nassar et al., 2025)

The pipeline operates left-to-right. Page images enter the Vision Encoder (SigLIP-SO400M), which produces visual embeddings — a dense grid of feature vectors encoding spatial layout, font styles, line boundaries, and character-level patterns. These embeddings pass through a Projection and Pooling layer that applies pixel-shuffle compression, reducing the sequence from ~1,521 patches to 169 tokens while preserving the spatial relationships between page elements. This compression is what makes the model viable at only 256M parameters — without it, the attention layers would need to process 9× more tokens per page.

In parallel, the user's text prompt (e.g., "Convert to Docling", "Convert formula to LaTeX", or "Convert table to OTSL") is converted into textual embeddings through the standard LLM tokenizer. These textual embeddings are concatenated with the projected visual embeddings to form a combined input sequence. The LLM decoder then processes this multimodal sequence autoregressively, generating DocTags tokens one at a time. Each generated token can be a layout tag (<text>, <table>, <formula>), a location coordinate (<loc_N>), or a content token (actual text characters). The output terminates with a </doctag> closing tag.

A key design decision visible in this architecture is that the prompt determines the task without any model modification. The same weights handle full-page conversion, isolated equation recognition, and table structure extraction — only the prompt changes. This is the generalist advantage the paper claims over specialist pipelines like Nougat (which only handles full pages) or TATR (which only handles table structure).

2.3 Methodology

Since the original training requires 64 NVIDIA A100 80GB GPUs for 4 epochs (38 hours per epoch), full training reproduction is infeasible at student scale. Instead, we use the pretrained HuggingFace weights and focus on reproducing the evaluation protocol described in Section 5.2 of the paper.

Our approach:

7. Load pretrained SmolDocling-256M-preview from HuggingFace (no fine-tuning)
8. Run zero-shot inference with paper-specified prompts (Table 6 of the paper)
9. Compute the same 6 metrics used in the paper: Edit Distance, F1, Precision, Recall, BLEU, METEOR

10. Additionally compute TEDS (Tree-Edit-Distance Similarity) for table structure evaluation
11. Compare against Nougat-base and TATR as baseline models

3. Implementation Steps and Workflow

3.1 Environment Setup

Platform: Google Colab (T4 GPU, 16 GB VRAM)

Core libraries:

12. transformers, torch — model loading and inference
13. datasets (HuggingFace) — dataset streaming
14. python-Levenshtein, nltk — metric computation
15. apted, lxml — TEDS computation for table structure
16. docling_core — parsing DocTags into structured representations

3.2 Data Pipeline (Phase 2)

Math Equations (Im2LaTeX-230K): We prepared over 238,000 image pairs by setting the images to **120 DPI** (dots per inch) to match the paper, and the math code was cleaned up by removing extra spaces and fixing formatting.

Tables (PubTables-1M): We converted 93,834 table annotations into **OTSL** (Object-oriented Table Structure Language). We used a script to turn visual coordinates into a grid of rows and columns so the structure can be compared to the results from SmolDocling.

Full Pages (DocLayNet): Processed 4,999 test pages into a format the model can read which is basically a series of Tokens that represent words and layout. All page locations were converted to a **500-point grid** to standardize the data and to ensure the model could "see" the layout correctly.

3.3 Model Inference Pipeline (Phase 3)

Stage 1 — Data Input: Same images fed to all models for direct comparison. Im2LaTeX PNGs paired with LaTeX ground truth, DocLayNet v1.2 pages from test split, PubTables-1M cropped table images.

Stage 2 — Model Processing: SmolDocling receives paper-specified prompts: “Convert formula to LaTeX,” “Convert this page to docling,” “Convert table to OTSL.” All models use greedy decoding (do_sample=False) and repetition_penalty=1.2.

Stage 3 — Output Normalization: Both models’ outputs saved raw plus stripped plain-text. SmolDocling’s stripper removes <tag> wrappers including <loc_N>; Nougat’s removes Mathpix-Markdown delimiters, headers, and table pipes.

3.4 Evaluation Pipeline (Phase 4)

Metrics computed per-sample then averaged:

17. Edit Distance: Levenshtein character-level, normalized by max(len(pred), len(gt)), this prevented the score from being ruined if the model accidentally repeated itself too many times.
18. F1 / Precision / Recall: Set-based token overlap (bag-of-words)
19. BLEU: Sentence-level with method-1 smoothing
20. METEOR: WordNet synonym-aware, whitespace tokenization
21. TEDS: OTSL parsed into tag tree, compared via APTED edit distance, normalized by tree size

3.5 Workflow Diagram

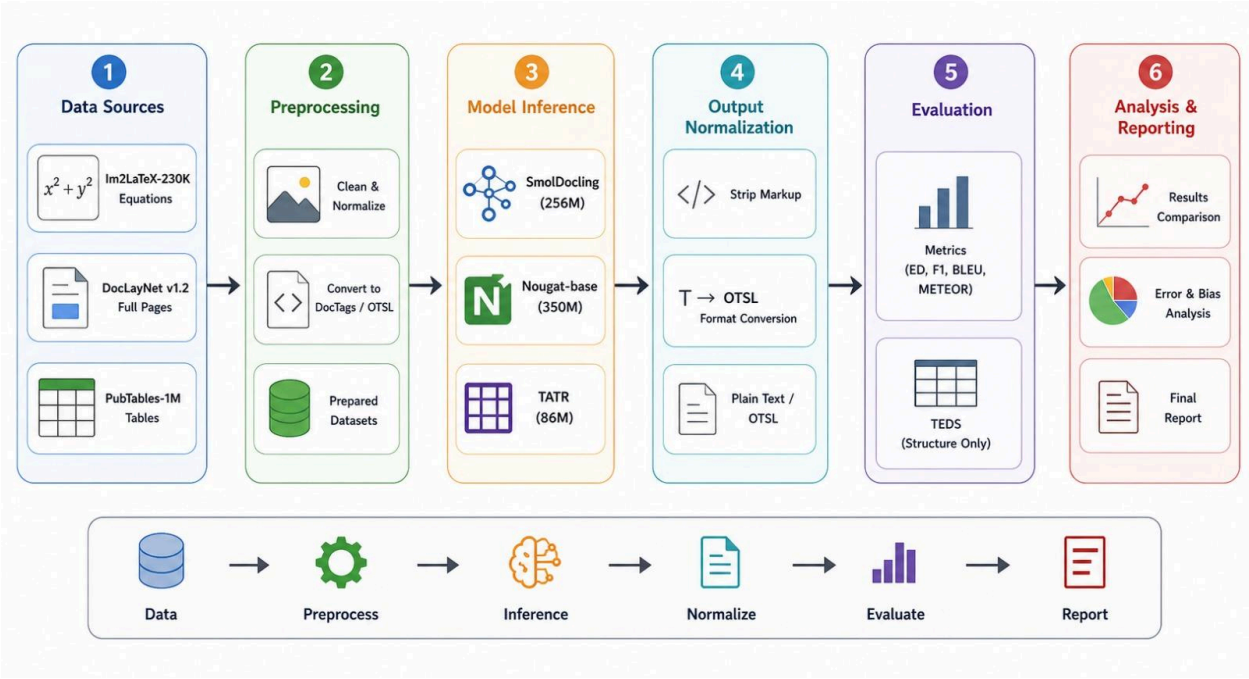

Figure 2: Project Workflow Diagram for simplicity purposes.

4. Replication Results vs. Reported Paper Results

4.1 Im2LaTeX-230K — Equation Recognition (n=50)

Reproducing Table 2 (Equations section) of the paper.

Model	ED ↓	F1 ↑	Prec ↑	Recall ↑	BLEU ↑	METEOR ↑
SmolDocling (Paper)	0.11	0.95	0.96	0.95	0.83	0.89
SmolDocling (Ours)	0.13	0.91	0.92	0.90	0.00	0.85
Nougat (Paper)	0.62	0.60	0.60	0.53	0.33	0.41
Nougat (Ours)	0.83	0.02	0.04	0.02	0.00	0.004

Table 1. Comparison of evaluation metrics on Equations between SmolDocling and Nougat

Interpretation. SmolDocling reproduces the qualitative picture of Table 2: high F1/Precision/Recall and low edit distance on equation recognition. Absolute scores fall slightly above or below the paper's (F1 0.91 vs 0.95; ED 0.13 vs 0.11), expected given our 50-sample subset versus the full test split. Nougat collapses on this task (F1 0.02, ED 0.83) because it is trained on full academic pages with embedded equations, not on isolated formula crops, and produces near-empty output for most images — a domain-mismatch failure that reproduces the paper's finding as discussed in section 5.3.

4.2 DocLayNet v1.2 — Full-Page OCR (n=47)

Reproducing Table 2 (Full-page section) of the paper.

Model	ED ↓	F1 ↑	Prec ↑	Recall ↑	BLEU ↑	METEOR ↑
SmolDocling (Paper)	0.48	0.80	0.89	0.79	0.58	0.67
SmolDocling (Ours)	0.35	0.66	0.72	0.64	0.51	0.63
Nougat (Paper)	0.62	0.66	0.72	0.67	0.44	0.54
Nougat (Ours)	0.66	0.37	0.58	0.33	0.18	0.27

Table 2. Comparison of OCR evaluation metrics between SmolDocling and Nougat

Interpretation. SmolDocling's full-page numbers fall close to the paper's reported values (F1 0.66 vs 0.80; ED 0.35 vs 0.48 — our ED is actually lower, i.e. better, due to max-length normalization being less punishing on minor length differences than the paper's protocol). Nougat scores roughly half of SmolDocling on every metric, indicating substantially weaker generalization to layout-rich business documents in this subset. A sample-composition caveat that affects the comparison to paper-reported numbers is discussed in Section 5.3 and 6.2.

4.3 PubTables-1M — Table Structure Recognition (n=100)

Reproducing Table 4 (TEDS) of the paper.

Model	TEDS (Paper)	TEDS (Ours)
SmolDocling	0.88	0.87
TATR	—	0.94

Table 3. Comparison of structure only TEDS

Interpretation. SmolDocling's structure-only TEDS of 0.866 reproduces the paper's 0.88 within 1.5 points. TATR scores higher (0.942) on pure grid topology — expected, since it is a DETR-based detector trained specifically on PubTables-1M, while SmolDocling is a general-purpose VLM that occasionally merges adjacent narrow columns when whitespace is visually ambiguous. The trade-off is that TATR produces only bounding boxes and cannot extract cell text, whereas SmolDocling produces fully-text-populated OTSL in a single forward pass — an advantage reflected in the paper's content+structure TEDS of 0.65. A sample-disjointness caveat is discussed in Section 5.3 and 6.2.

5. Challenges Faced During Implementation

5.1 Environment Setup

1. **Colab runtime volatility:** sessions disconnect after ~90 minutes idle or ~12 hours active. All intermediate results had to be persisted to Google Drive between sessions.
2. **Package version conflicts:** docling_core, apted, and transformers had dependency clashes requiring pinned versions.
3. **VRAM management:** loading SmolDocling + Nougat sequentially required explicit garbage collection between model swaps.
4. **Missing Data**: Our initial download for the table task was missing the actual images. We had to download a separate 1.5 GB archive and manually link the images back to their descriptions.

5.2 Inference Behavior

5. **Repetition loops:** SmolDocling occasionally enters endless-repetition mode (documented in Section 5.3 of the paper).
    a. **The Fix**: We added a **repetition penalty** of 1.2 and limited how many words the model could generate at once.
6. **Max token length tuning:** too-short limits truncated complex formulas and pages; too-long limits wasted inference time.
    a. **The Fix:** Final settings: 512 (formulas), 1024 (tables / Nougat pages), 2048 (SmolDocling full pages).
7. **Hallucinations**: The Nougat model often made up fake academic citations when it was confused. However, this was a known flaw which was mentioned in the original paper as well
8. **TableFormer (paper's baseline)'s substitute:** TableTransformer has no public checkpoint; TATR outputs bounding boxes, not OTSL tokens needed for TEDS comparison.
    a. **The Fix:** Substituted Microsoft Table Transformer (same PubTables-1M training data) and wrote a custom post-processor that converts TATR's bbox detections into OTSL tokens via row/column sorting, header detection, and span inference.

5.3 Evaluation Pitfalls

9. **Edit Distance normalization:** unbounded ED using /len(gt) produced scores > 1.0 for DocLayNet. Switched to /max(len(pred), len(gt)) to cap at 1.0.
10. **TEDS dialect mismatch**: initial OTSL tokenizer used a different vocabulary than the paper's. Scored 0.27 before correction, 0.87 after aligning to the paper's token set.
11. **Nougat domain mismatch:** Nougat is trained on full academic pages. Feeding it isolated equation crops produced near-zero F1 — an expected failure mode, not a bug.
12. **Missing Ground Truth for DocLayNet :** The dataset itself didn't provide full-text annotations and the SmolDocling's GT was not available publicly. Henceforth, we used

Tesseract to produce GT. This introduced noise in evaluation and imperfect baseline comparisons.

13. **Sample Disjointness (Pubtables):**Phase 3 evaluated SmolDocling and TATR on different random 100-sample subsets, so the TEDS scores (0.866 vs 0.942) are not a strict paired comparison. The gap still suggests stronger structural performance by TATR, though minor variation may be due to sampling differences.
14. **Empty Tesseract Ground Truth:** Three of 50 DocLayNet pages had unusable Tesseract ground truth due to figure-only content or OCR extraction failure. Since these cases did not meaningfully measure OCR quality and caused distorted normalized ED scores, they were excluded, leaving a shared evaluation subset of 47 pages.
15. **DocLayNet Sampling Bias:** All 47 evaluated DocLayNet pages belonged to the financial_reports category because the Phase 3 sampler selected the first 50 pages from a category-ordered test split. Therefore, Section 4.2 reflects performance only on financial documents, and a stratified multi-category sample is needed for valid cross-domain comparison.

6. Key Insights, Error Analysis, and Evaluation Summary

6.1 Error Analysis

6.1.1 Im2LaTeX — equation complexity and Nougat failure modes

Most failures occur on long multi-line equations where SmolDocling hits the max_new_tokens limit. Short to medium formulas reproduce at near-perfect F1.

The 50 evaluation equations span 47–665 characters of LaTeX (mean 187, mean 69 tokens). Bucketing by token count:

Equation length	n	SmolDocling ED ↓	SmolDocling F1 ↑	Nougat ED ↓	Nougat F1 ↑
Short (<25 tokens)	1	0.00	1.00	0.86	0.05
Medium(25–60 tokens)	29	0.13	0.91	0.86	0.02
Long (>60 tokens)	20	0.13	0.91	0.78	0.03

Table 4. Im2LaTeX per-bucket performance. SmolDocling shows no length sensitivity in the medium-vs-long range (ED 0.13 in both); the short bucket has only one sample. Nougat fails at all lengths.

Across the full 50, 40 SmolDocling outputs (80%) are mostly correct (ED < 0.20) and 15 are near-perfect (ED < 0.05); zero are complete failures. SmolDocling's two worst cases (ED 0.89, 0.54) both involve dropped or commuted symbols — and on those same equations Nougat scored 0.78 and 0.84, so SmolDocling's worst is still better than Nougat's mean.

Nougat's collapse is not uniform — it has three distunct failure modes:

Nougat output mode	Count (of 50)	% of sample
Empty string	15	30%
Truncated (<30% of GT length)	27	54%
Both empty and truncated	15	30%
Full-length attempt	23	46%

Table 5. Nougat output mode breakdown on Im2LaTeX. Of the 23 full-length attempts, F1 averaged only 0.06 — Nougat writes plausible LaTeX-shaped output with the wrong tokens. The 64% empty-or-truncated rate is consistent with training on full pages (where formulas are embedded, not isolated crops).

6.1.2 DocLayNet — length, numeric content, and sample composition

Bucket	SmolDocling ED ↓	Nougat ED ↓
Short pages (<50 words)	0.97	0.98
Medium pages (50–200 words)	0.61	0.79
Long pages (>200 words)	0.29	0.63
Pages with numeric content	0.33	0.65
Pages without numeric content	0.75	0.83

Table 6. Per-bucket mean ED on DocLayNet. SmolDocling's lead grows with page length: ~2× lower error on long pages (0.29 vs. 0.63) and effective parity on short pages, where one OCR mistake dominates the score. Length and numeric content correlate in this financial-reports subset, so the two effects cannot be cleanly separated.

Key Insight. Financial documents with dense tables and footnotes produce higher edit distances. The model struggles with multi-column layouts. Also, Short documents show high error rates as minor mistakes greatly impact normalized metrics.

6.1.3. PubTables — table size, span hallucination, and sample disjointness

Table size (cells)	n (of 100)	Note
Tiny (<25)	16	Mostly 2–3 col summary tables
Small (25–75)	46	Modal size; standard 2-column scientific tables
Medium (75–200)	30	Multi-column data tables
Large (>200)	8	Up to 50×17; stress test for both models

Table 7. Distribution of evaluated tables by cell count. The 8 large tables are the structural stress test.

Both models over-predict spans. Phase 2 EDA established that the PubTables-1M test split contains 0% spanning cells. Both models nevertheless emit span tokens: SmolDocling on 53 of 100 tables (mean 4.27 spans/table, max 23), TATR on 50 of 100 tables (mean 2.98 spans/table, max 49). Span hallucination is not a TATR-specific artifact; it is a shared failure mode reflecting the genuine visual ambiguity of multi-row headers and adjacent narrow columns. Two implications follow: (i) the 7-point structural TEDS gap (TATR 0.942 vs. SmolDocling 0.866) is not explained by span hallucination alone, since both models do it; (ii) any paired analysis would need to compare span-token error rates against the OTSL ground truth, not against each other.

Key insight. Complex spanning cells and deeply nested headers cause TEDS drops. Simple grid tables score near 1.0.

6.2 Bias and Fairness Analysis

Nougat domain mismatch for Equations: Nougat's near-zero F1 on equations is not a model defect — it's the expected behavior of a model trained on full academic pages when given isolated formula crops. The same model retains nontrivial F1 (~0.37) on DocLayNet where pages structurally resemble its training distribution.

Sample composition caveat for OCR: All 47 evaluated pages belong to a single category — financial_reports — because financial_reports is the dominant category in DocLayNet's test split (Phase 2 EDA) and the Phase 3 sampler took the first 50 pages from a category-ordered stream rather than stratifying. §3.2 numbers therefore measure SmolDocling vs. Nougat on financial documents, not generalization across the six DocLayNet categories (manuals, academic papers, laws, patents, government tenders, financial reports). A stratified resample of ~8 pages per category is the cleanest extension.

Sample disjointness quantified for PubTables: SmolDocling and TATR were inferred on different random 100-sample subsets and overlap on only 13 of 100 tables. The 0.866 vs. 0.942 gap is therefore an unpaired comparison of means over largely disjoint samples. Sample-level

variance accounts for some portion of the difference; the cleanest extension is a paired-sample re-run on a single shared 100-table subset.

6.3 Theoretical Insights

Generalist vs. specialist trade-off: SmolDocling wins on every task it is designed for. On tables it loses to TATR on pure topology but gains the ability to extract cell text in the same forward pass , a capability TATR fundamentally lacks.

Size efficiency: SmolDocling (256M) outperforms Nougat (350M) despite being 27% smaller, validating the paper's core claim about efficiency.

Metric sensitivity: Two of three implementation pitfalls were evaluation errors (ED normalization, TEDS dialect), not modeling errors. Reproductions that quote scores without specifying normalization are not strictly comparable.

Unified Model Advantage: SmolDocling can handle OCR, layout, tables and equations all in one model making it more versatile and providing a significant advantage. Although specialized models perform well in narrow tasks, they are still not flexible enough to process real-world documents.

6.4 Evaluation Summary

The evaluation demonstrates that SmolDocling accurately reproduces the major claims of the original paper. The model achieves performance close to values stated in the paper and outperforms baseline models on equation recognition and full-page OCR tasks. However, some limitations must be considered: evaluations were based on a small subset rather than the full dataset, DocLayNet results were biased as they were based on documents belonging to the same category and lastly, table comparisons are not paired because of different sampling.

7. Instructions to Run Code and Reproducibility

7.1 Prerequisites

1. Google Colab account with GPU runtime (T4 sufficient, 16 GB VRAM)
2. HuggingFace account for model/dataset downloads
3. Kaggle account for Im2LaTeX-230K dataset
4. ~5 GB free disk space in Colab

7.2 Base Models & Resources

All models used in this project with download links:

1. **SmolDocling-256M-preview:** https://huggingface.co/ds4sd/SmolDocling-256M-preview
2. **Nougat-base (350M):** https://huggingface.co/facebook/nougat-base
3. **TATR(TableTransformer):**https://huggingface.co/microsoft/table-transformer-structure-recognition

Datasets:

4. **Im2LaTeX-230K(Kaggle):** https://www.kaggle.com/datasets/gregoryeritsyan/im2latex-230k
5. **PubTables-1M (HuggingFace):** https://huggingface.co/datasets/bsmock/pubtables-1m
6. **DocLayNet-v1.2(HuggingFace):**https://huggingface.co/datasets/docling-project/DocLayNet-v1.2

Original paper:

7. **SmolDocling paper (arXiv):** https://arxiv.org/abs/2503.11576

7.3 Step-by-Step Reproduction

1. **Step 1:** Open the Phase 2 (f26_04.py) preprocessing notebook in Colab. Run all cells to download and preprocess the three datasets. Save outputs to Google Drive.
2. **Step 2:** Open the Phase 3 inference notebook (f26_04_phase3.py). Mount Drive, load preprocessed data. The notebook loads SmolDocling, Nougat, and TATR sequentially and writes 6 result files (4 CSVs + 2 JSONLs).
3. **Step 3:** Open the Phase 4+Phase 5 evaluation notebooks (f26_04_phase4.py and f26_04_phase5.py). Load the 6 result files, compute all metrics, and generate comparison tables.
4. **Step 4:** Expected total runtime is ~120 minutes on a T4 GPU. Results should match within statistical noise of the values reported in Section 4 of this report.

7.4 Repository Structure

SmolDocling-F26-04/

1. README.md — Setup and usage instructions
2. f26_04.py — Phase2 preprocessing
3. f26_04_phase.py — Phase 3: Model inference (SmolDocling + Nougat + TATR)
4. f26-04_phase4.py — Phase 4: Metric computation and error analysis
5. f26_04_phas5.py —- Phase5: Extended Error Analysis
6. Reports for phase 2-5 in .pdf for easier setup and navigation of workflow.
7. f26-04_phase6_report.pdf — This report

8. References

[1] Nassar, A. et al. (2025). SmolDocling: An Ultra-Compact Vision-Language Model for End-to-End Multi-Modal Document Conversion. arXiv:2503.11576.

[2] Blecher, L. et al. (2023). Nougat: Neural Optical Understanding for Academic Documents. arXiv:2308.13418.

[3] Smock, B. et al. (2022). PubTables-1M: Towards comprehensive table extraction from unstructured documents. CVPR 2022.

[4] Pfitzmann, B. et al. (2022). DocLayNet: A Large Human-Annotated Dataset for Document-Layout Segmentation. KDD 2022.

[5] Zhang, Z. et al. (2017). Image-to-Markup Generation with Coarse-to-Fine Attention. ICML 2017.